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Section 5.6

5.6–3

The sample mean $\bar{X}$ of size $n = 36$ from an exponential distribution with mean 3 has

$$\mu_{\bar{X}} = 3, \qquad \sigma_{\bar{X}} = \frac{3}{\sqrt{36}} = \frac{1}{2}.$$

Using the CLT,

$$P(2.5 \le \bar{X} \le 4) = P\left(\frac{2.5-3}{0.5} \le Z \le \frac{4-3}{0.5}\right) = P(-1 \le Z \le 2) \approx 0.97725 - 0.15866 = 0.8186.$$

$$\boxed{P \approx 0.819}$$

5.6–5

Each $X_i \sim \chi^2_1$.

(a) The sum

$$Y = \sum_{i=1}^{18} X_i \sim \chi^2_{18}.$$

(b) Approximation using CLT: Mean and variance of each: $\mu = 1,\ \sigma^2 = 2$. For the sum:

$$E[Y] = 18, \qquad Var(Y) = 36, \qquad \sigma_Y = 6.$$

For 9.390:

$$z = \frac{9.390 - 18}{6} = -1.435, \quad P \approx 0.0755.$$

For 34.80:

$$z = \frac{34.80 - 18}{6} = 2.8, \quad P \approx 0.9974.$$

The CLT overestimates the lower tail (true = 0.05) and is closer on the upper tail (true = 0.99).

5.6–6

The pdf is $f(x) = 1 - \frac{x}{2}$ for $0 \le x \le 2$.

(a) Compute the mean:

$$\mu = \int_0^2 x\left(1 - \frac{x}{2}\right) dx = 2 - \frac{4}{3} = \frac{2}{3}.$$

Compute $E[X^2]$:

$$E[X^2] = \int_0^2 x^2\left(1 - \frac{x}{2}\right) dx = \frac{8}{3} - 2 = \frac{2}{3}.$$

Thus

$$\sigma^2 = E[X^2] - \mu^2 = \frac{2}{3} - \frac{4}{9} = \frac{2}{9}.$$

(b) For $\bar{X}$ with $n = 18$:

$$\sigma_{\bar{X}}^2 = \frac{2/9}{18} = \frac{1}{81}, \quad \sigma_{\bar{X}} = \frac{1}{9}.$$

Standardizing interval:

$$P\left(\frac{2}{3} \le \bar{X} \le \frac{5}{6}\right) = P\left(0 \le Z \le \frac{\frac{1}{6}}{1/9}\right) = P(0 \le Z \le 1.5).$$

$$P \approx 0.93319 - 0.5 = 0.433.$$

$$\boxed{P \approx 0.433}$$

5.6–12

Interarrival time $T \sim \Gamma(\alpha = 3, \theta = 2)$. Selling 10 tickets: sum of 10 gammas:

$$S = \sum_{i=1}^{10} T_i \sim \Gamma(30, 2).$$

(a) Probability sold out within 60 minutes:

$$P(S \le 60) = \int_0^{60} \frac{1}{\Gamma(30)2^{30}} x^{29} e^{-x/2}\, dx.$$

(b) CLT approximation:

$$E[T] = 6, \qquad Var(T) = 12.$$

$$E[S] = 60, \qquad Var(S) = 120, \qquad \sigma_S = \sqrt{120}.$$

$$P(S \le 60) \approx P(Z \le 0) = 0.5.$$

$$\boxed{P \approx 0.50}$$

Section 5.7

5.7–2

$X \sim \text{Binomial}(25, 0.2)$.

(a) Exact:

$$P(3 \le X \le 8) = \sum_{k=3}^{8} \binom{25}{k} (0.2)^k (0.8)^{25-k} \approx 0.855.$$

note: since the problem specifically asks to use table in appendix. I also used the table values:

$$P(X \le 8) - P(X \le 2) \;=\; 0.9532 - 0.0982 \;=\; 0.855.$$

This agrees with the computed probability uses the appendix.

(b) Normal approximation with continuity correction: Mean = 5, variance = 4, $\sigma = 2$.

$$P(3 \le X \le 8) \approx P(2.5 < X < 8.5).$$

$$z_{\text{low}} = \frac{2.5 - 5}{2} = -1.25, \qquad z_{\text{up}} = \frac{8.5 - 5}{2} = 1.75.$$

$$P \approx \Phi(1.75) - \Phi(-1.25) = 0.95994 - 0.10565 = 0.8543.$$

$$\boxed{0.854 \text{ (approx)}}$$

5.7–7

$X \sim \text{Poisson}(49)$.

Normal approximation: mean 49, variance 49, $\sigma = 7$.

We want $45 < X < 60$. With continuity correction:

$$P(45 < X < 60) \approx P(45.5 < X < 59.5).$$

$$z_{\text{low}} = \frac{45.5 - 49}{7} = -0.5, \qquad z_{\text{up}} = \frac{59.5 - 49}{7} = 1.5.$$

$$P \approx \Phi(1.5) - \Phi(-0.5) = 0.93319 - 0.30854 = 0.62465.$$

$$\boxed{P \approx 0.625}$$